

Quantify how well the forward surrogate generalizes to design parameters it has never seen before

Probe the in-between regions of SQuADDS by sampling uniformly inside the training cloud and binning by proximity.

1. Sample uniformly

Generate 50,000 random design parameter sets:
uniform inside the convex hull of the training data (interpolation only).



2. Measure NN distance

For each random point:
compute d_{NN} = Euclidean distance to its nearest training sample in scaled $[0, 1]$ space.



3. Bin and select

Split the d_{NN} range into 10 equal-width bins (close $\rightarrow$ far):
draw 10 random points per bin
 $\rightarrow$ 100 total for EM validation.

Surrogate evaluates all 50,000 samples in seconds.
100 selected across distance bins sent to the EM solver for validation.